

Multi-Agent Financial Research System

Milestone 1: Research Agent Design & Logical Architecture

1. Overview of the Research Agent

The Research Agent (Agent #5) is responsible for answering user queries using a Retrieval-Augmented Generation (RAG) pipeline. Instead of relying on general knowledge, the agent retrieves only the most relevant information from uploaded financial documents such as annual reports, 10-K filings, earnings transcripts, and investor presentations. This ensures that every response is grounded in verified source documents, minimizing hallucinations and improving the reliability of AI-generated insights.

The agent also provides source citations for every factual statement. Each citation includes the company name, document section, and page number, allowing users to verify the information directly from the original source.

2. Objectives

The Research Agent is designed with the following objectives:

- Answer financial queries accurately using document-based evidence.
 - Retrieve semantically relevant document chunks from the Vector Database.
 - Generate responses that are grounded only in retrieved context.
 - Prevent hallucinations by restricting the language model from using external knowledge.
 - Provide clear and transparent source citations for every response.
 - Improve the reliability and trustworthiness of AI-assisted financial research.
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3. Logical Architecture & Workflow

The Research Agent follows a structured workflow to process every user query:

Step 1: User Query

The user submits a financial question through the conversational interface.

Step 2: Vector Database Retrieval

The query is converted into embeddings and matched against the Vector Database to retrieve the most relevant document chunks.

Step 3: Context Collection

The retrieved document chunks, along with their metadata (Company Name, Section, and Page Number), are collected.

Step 4: Prompt Construction

The retrieved context and the user's query are combined into a structured prompt that will be sent to the Language Model.

Step 5: Grounded Response Generation

The Language Model generates an answer strictly using the retrieved context while following predefined system instructions.

Step 6: Citation Generation

The final response is returned together with source citations so that users can verify every fact presented.

4. Prompt Grounding Strategy

To ensure factual correctness, the Research Agent uses a strict prompt grounding strategy.

System Instructions

- Answer only using the information provided in the retrieved context.
- Do not assume or infer information that is not explicitly present.
- Do not use external knowledge.
- If the required information is unavailable, return:

"Information not found in source documents."

- Every factual statement must include its corresponding citation in the following format:

[Company | Section, Page Number]

This approach minimizes hallucinations and ensures that every response is traceable to the original financial document.

5. Retrieval-Augmented Generation (RAG) Pipeline

The Research Agent implements a Retrieval-Augmented Generation (RAG) workflow consisting of the following stages:

1. User submits a financial query.
2. Query embeddings are generated.
3. Similar document chunks are retrieved from the Vector Database.
4. Retrieved context is combined with metadata.
5. A grounded prompt is constructed.
6. The prompt is passed to the Language Model.
7. The Language Model generates a response.
8. The final answer is displayed with appropriate source citations.

6. Key Features

The proposed Research Agent provides several important features:

- Retrieval-Augmented Generation (RAG)
 - Semantic search using vector embeddings
 - Context-aware response generation
 - Hallucination prevention
 - Source citation support
 - Financial document understanding
 - Transparent and verifiable answers
 - Scalable architecture for future integration with production databases and LLM services
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7. Advantages

The proposed architecture offers several advantages:

- Improves the factual accuracy of AI-generated responses.
 - Ensures transparency through document citations.
 - Reduces misinformation and hallucinations.
 - Enables users to verify responses directly from source documents.
 - Provides a scalable foundation for enterprise-grade financial research applications.
 - Supports future integration with real-time vector databases and large language models.
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8. Future Enhancements

The current milestone focuses on the logical design of the Research Agent. Future development phases will include:

- Integration with PDF document parsers.
- Automatic document chunking and embedding generation.
- Connection to a production Vector Database.
- Integration with Large Language Models through APIs.
- Multi-document retrieval and ranking.
- Advanced citation formatting and response evaluation.
- Performance optimization for large-scale financial datasets.

9. Conclusion

The Research Agent architecture provides a reliable and scalable solution for document-grounded financial question answering. By combining Retrieval-Augmented Generation with strict prompt grounding and citation-based responses, the system minimizes hallucinations while ensuring that every generated insight is supported by verifiable evidence from the uploaded financial documents. This milestone establishes the logical foundation for implementing a complete AI-powered financial research assistant in the subsequent phases of the project.